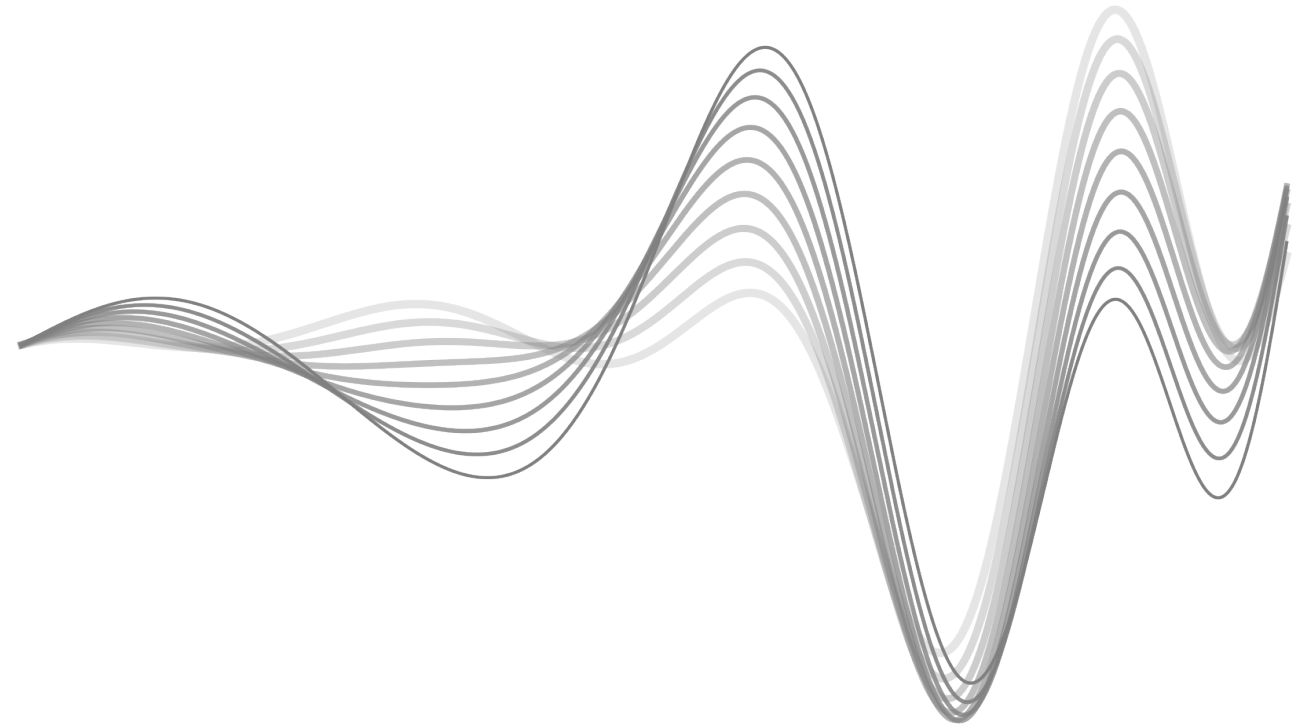




Welcome!





Who is in the room?

Choose Your Wall



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

	[1]	[2]	[3]	[4]
1. Residence / continent:	EU /	Asia /	Middle-East /	Other
2. Academic level:	undergrad /	grad /	PhD /	Other
3. Prior knowledge DL:	little /	some /	more /	professional
4. Python / Pytorch skills:	little /	some /	advanced /	expert

Merten Stender



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

- 2023: **Professor** (W1)
Cyber-Physical Systems in Mechanical Engineering
Technische Universität Berlin



- 2020: **PhD** from Hamburg University of Technology
Supervisor: Prof. Norbert Hoffmann
Data-Driven Techniques for the Nonlinear Dynamics of Mechanical Structures



- 2019: **Founder** of TensorDynamic GmbH
Machine Learning Consultancy and Software Engineering



- Research stays: Imperial College London (Rolls-Royce VUTC),
University of Technology Sydney, Sandia Laboratories,
Albuquerque/USA, AUDI AG Ingolstadt

Imperial College
London



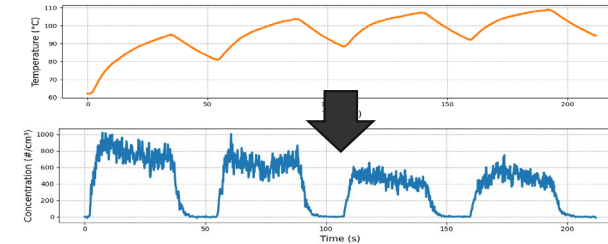
Michal Brzosko

- PhD candidate (2nd year)

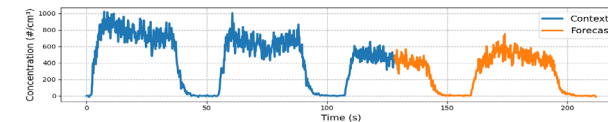


Cyber-Physical Systems
in Mechanical Engineering TU Berlin

Seq2seq translation



Time series forecasting



Explainable AI

Applications:

- Automotive brake systems
- Airborne particulate matter
- Virtual/ soft sensing

[DFG 545840087](#): **SLOPE**
Partner: University of Lille



- Transformer
- Neural Operators
- PINNs

PyTorch





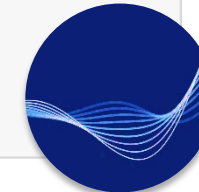
Applied Machine Learning for Engineering Systems

- ML & DL for Digital Twins
- Physics-informed Learning
- Explainable AI
- Efficient and green ML



Dynamical systems and complex dynamics

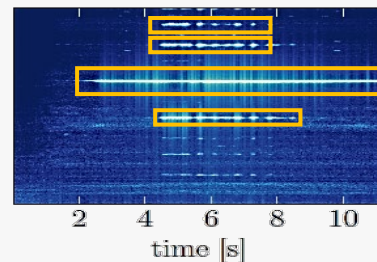
- Advanced signal processing
- Learning for dynamics
- Time series forecasting
- Alternative Computing



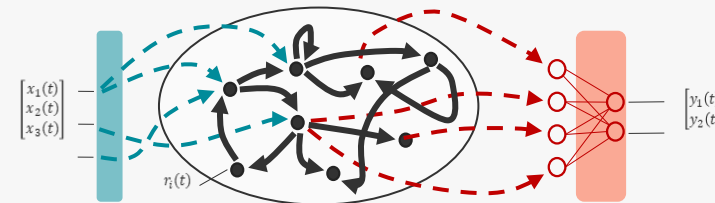
Prof. Dr.-Ing. Merten Stender

Research & Applications

Predictive maintenance & computer vision



Reservoir Computing



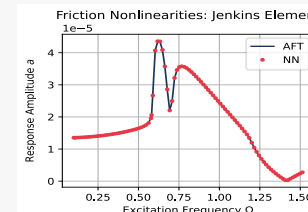
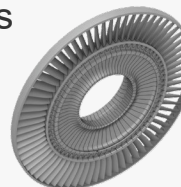
Team

- 5 PhDs
- 2 Postdocs
- 3 Students

Nonlinear ocean wave reconstruction and forecasting



Nonlinear structural dynamics



Funding

- DFG
- BMW, BMWK
- A. v. Humboldt
- EU

*June 2026



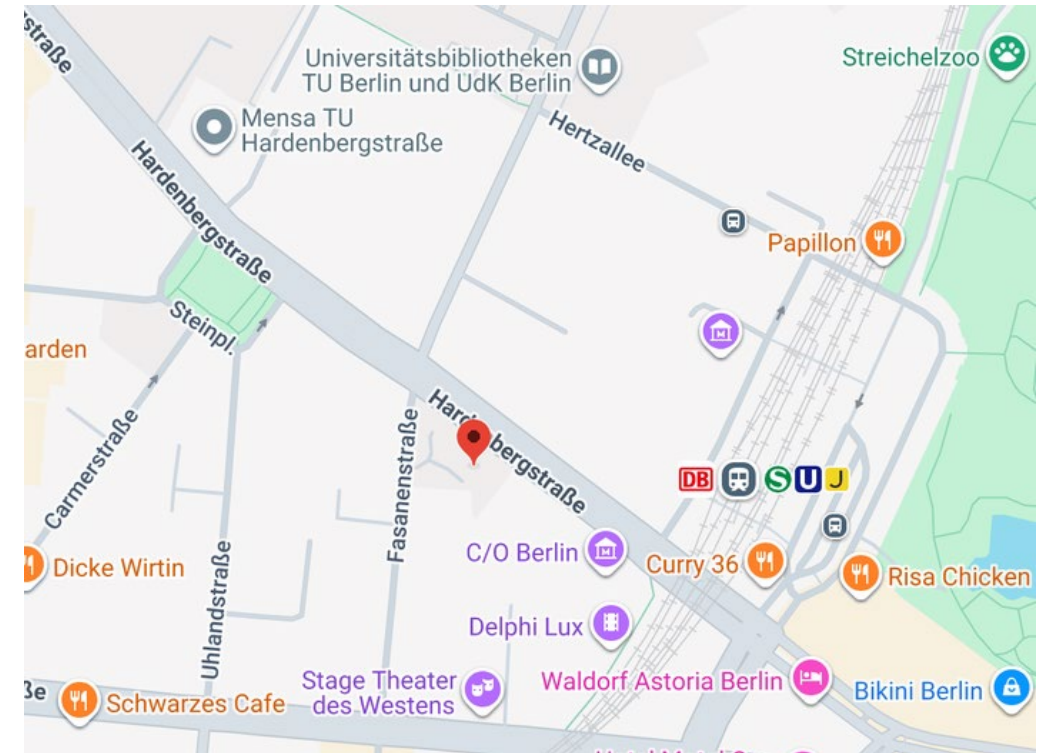
First-day survival guide

Organization



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

- **Room and daily times:** HBS 010
 - Slot 1: 09:00-10:30
 - Slot 2: 10:45-12:15
 - Slot 3: 13:15-14:45
 - Slot 4: 15:00-16:30
- **Breaks:** small breaks during morning and afternoon, lunch break 12:15-13:15
- **Wi-fi:** Free Berlin Wifi; eduroam (recommended)
- **Communication channel:** email to [merten.stender](mailto:merten.stender@tu-berlin.de) and michal.brzosko@tu-berlin.de
- **Photo policy:** anyone not OK with being on pictures? LinkedIn welcome



Attendance/certificate expectations



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

- The course awards **3 ECTS** (minimum total workload of 75 hours, incl. 15hrs cultural activities)
- **Attendance** (43,5 hrs)
 - Students must attend at **least 80% of all class sessions** to be eligible to pass.
 - Attendance must be recorded for every session, not merely once per day.
- **Workload, certificate and grading**
 - **Homework** must be assigned so that total work corresponds to the ECTS workload.
 - The course must contain an **assessable final test or project**.
 - Individual final grade using the German system (1.0 – 4.0)
- **Assessment**
 - Mid term presentation (Friday, 21 August)
 - Final presentation (Friday, 28 August)



The 2-week learning contract

Schedule



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

Week 1

Day	Slot 1	Slot 2	Slot 3	Slot 4
	09:00-10:30	10:45-12:15	13:15-14:45	15:00-16:30
Mon, Aug 17	Orientation/Welcome	Orientation/Welcome	Lecture (MS): Introduction to Deep Learning (time series data, tasks, application domains) 13:45-15:45	
Tue, Aug 18	Lecture (MS): Deep Learning basics	Lecture (MS): Recurrent Neural Networks (RNN, LSTM, GRU)	Lecture (MB): Transformer (self attention, positional encoding, encoder/decoder)	Practical (MB): Intro to Projects (dataset overview, group matching)
Wed, Aug 19	Lecture (MB): PyTorch4TS 101 (first PyTorch coding, LSTM)	Practical (MB): Build a first prototype	Cultural Program *	
Thu, Aug 20	Lecture (MB): Model Training & Evaluation (early stopping, data splitting, metrics)	Practical (MB): Build models	Lecture (MS): Hyperparameter Optimization (grid/line search, AutoML, k fold CV)	Practical (MB): Build models
Fri, Aug 21	Practical (MB): Build models (final prep, scoring, evaluation)	Assessment (MS+MB): Mid Term Presentation (baseline model: CNN, RNN, or Transformer)	Cultural Program *	
Sat, Aug 22	Cultural Program *			

basics of deep learning

classic neural architectures
for time series

basics of python and pytorch
practical phases @you

modeling competences

mid term presentation
(baseline modeling approach)

Schedule



Week 2

Day	Slot 1	Slot 2	Slot 3	Slot 4
Mon, Aug 24	Lecture (MS): Advanced Concepts ((x/s/m)LSTM)	Lecture (MS): Advanced Concepts II (Mamba, S4)	Practical (MB): Adapting an xLSTM to a given problem	Cultural Program *
Tue, Aug 25	Lecture (MB): Students build advanced models	Lecture (MS): Students build advanced models	Lecture (MS): Q&A and advanced concepts (hardware/GPU considerations)	Practical (MS): Autoregressive Inference (error accumulation, stability)
Wed, Aug 26	Lecture (MS): Foundation Models (self supervised learning, probabilistic forecasting)	Practical (MB): Guided Assignment Work	Cultural Program *	
Thu, Aug 27	Practical (MB): Guided Assignment Work	Practical (MB): Guided Assignment Work	Practical (MS): Additional Theoretic Input (interpretability, on request)	Practical (MS): Guided Assignment Work
Fri, Aug 28	Assessment (MB+MS): Presentation of Results	Assessment (MB+MS): Presentation of Results	Assessment (MB+MS): Presentation of Results	Orientation/Farewell (16:00)

advanced neural architectures
for time series

foundation models

final presentation
(baseline modeling approach)

* Activity times may vary; details announced during the program.



Repository walkthrough



GitHub repository

[Link](#)



Cyber-Physical Systems
in Mechanical Engineering TU Berlin

DL_TimeSeries_SummerSchool Private

Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Add file Code

mertstend Update schedule with instructor initials and times 118c841 · 2 weeks ago 55 Commits

projects	Fix LaTeX formatting in dataset description	2 weeks ago
teaching_material	Add session_07_PyTorch_Transformer	2 weeks ago
.gitignore	Initial commit	last month
PREREQUISITES.md	Update PREREQUISITES.md	2 weeks ago
README.md	Update schedule with instructor initials and times	2 weeks ago
RESOURCES.md	update resources	2 weeks ago

README

Deep Learning for Time Series Modeling: Summer School

TU Berlin | M. Stender & M. Brzosko

A hands on summer school covering deep learning methods for time series modeling: sequence to sequence architectures, forecasting, and data driven modeling of dynamical systems.

teaching_material/

Lecture slides and accompanying Jupyter notebooks. Notebooks can be run on Google Colab or locally.

About

Summer School at TU Berlin on Deep Learning for Time Series Modeling (M. Stender, M. Brzosko)

- Readme
- Activity
- Custom properties
- 0 stars
- 0 watching
- 0 forks
- Audit log

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Contributors 2

- michalbrzosko** Michal
- mertstend** Merten Stender



Questions?

5 min break